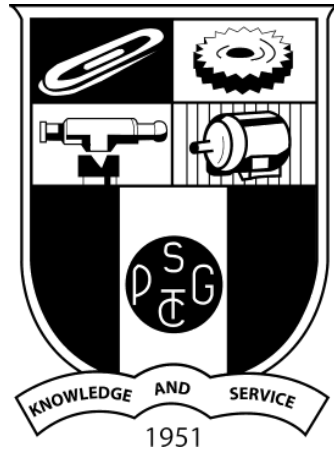


PSG COLLEGE OF TECHNOLOGY

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Department of Computer Science and Engineering

PROJECT WORK 1 ABSTRACT

**Automated Classroom Attendance from a Single Handheld
Sweep Video: A CPU-Oriented Benchmark of Face Detectors
under Real Classroom Conditions**

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Problem Statement:

Manual roll-calls waste valuable instructional time and are highly susceptible to proxy attendance. Although automated camera-based solutions exist, they are predominantly designed for idealised laboratory settings using curated, front-facing images, controlled lighting, and high-end GPU hardware. In real classrooms, the most practical capture device is a teacher's handheld phone, which introduces challenges such as panning-induced motion blur, varying camera angles, and rear-row distance gradients. Furthermore, because current systems rely on GPU-backed inference, they cannot run efficiently on unaccelerated commodity CPU hardware.

Abstract:

Manual roll-calls consume valuable instructional time and remain vulnerable to proxy attendance, motivating automated camera-based solutions. However, most existing systems are designed for controlled laboratory settings using curated front-facing images and GPU-accelerated processing. In real classrooms, teacher-owned handheld devices are the most practical capture method but introduce challenges such as motion blur from panning, uneven illumination, and small rear-row faces. Moreover, existing benchmarks primarily evaluate still images and rarely profile CPU-only edge hardware, while tracking-based methods require continuous fixed-camera monitoring that is impractical on commodity systems.

This project proposes an automated classroom attendance pipeline that processes a single handheld sweep video entirely on commodity CPU hardware. The video is temporally subsampled at 4 fps, and blurred frames are discarded using a variance-of-Laplacian sharpness filter. Face detection is performed on downscaled frames, while high-resolution ArcFace embeddings are extracted only from cropped facial regions to preserve rear-row detail. Global camera motion is compensated through homography estimation, after which tracklets are consolidated into unique student identities using agglomerative clustering before matching against a pre-enrolled class roster.

Unlike prior approaches, the proposed framework combines handheld video capture with embedding-based identity consolidation in a completely GPU-free pipeline. It extends static image-based systems by introducing video input, evaluating performance under realistic classroom variations (angle, distance, illumination, and occlusion), and benchmarking four face detectors (RetinaFace ResNet50/MobileNet0.25, YOLOv8-face, MTCNN, and YuNet) on CPU hardware. Evaluation uses an all-present baseline and asymmetric decision thresholds to minimise false absences, providing a practical, resource-efficient, and transparent classroom attendance solution.

References:

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